IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EIE PART II: MEng, BEng and ACGI

**COMPUTER ARCHITECTURE**

Wednesday, 13 May 10:00 am

Time allowed: 1:30 hours

**There are TWO questions on this paper.**

**Answer TWO questions.**

*Answer ALL questions.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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1 In the following question, the digits $C_3, C_2, C_1$ refer to the three least significant digits of your CID (College Identifier). For example, if your CID is shown on your swipe card as '00123456' then $C_3 = 4$, $C_2 = 5$, $C_1 = 6$.

a A CPU manufacturer's data-sheet quotes performance in terms of how many instructions are completed in one second.

i) What is the standard three- or four-letter acronym used to describe this metric, and what does it stand for?

ii) Give one advantage and one dis-advantage of this performance metric.

b A program is compiled and run in a system with a five-stage pipelined MIPS-1 CPU running at 80MHz. The CPU pipeline is identical to the one seen in lectures, and includes an instruction and data cache. Empirical tests show the system completes $(1 + C_1) \times 10^7$ instructions for each second the program runs.

i) State your value of $C_1$, and calculate the IPC for this program.

ii) Choose one of the following categories to describe the achieved IPC, and justify your choice:

A: The CPU is achieving optimal performance.

B: The program experiences a typical number of stalls.

C: The program causes a high number of cache misses.

D: The achieved IPC is impossible in a five-stage MIPS.

c Assume a memory system with an L1 and L2 cache connected to a DDR RAM. The hit time of the L1 cache is $h_{L1} = 1 + C_2$ cycles, and the miss-rate of the L1 cache is $m_{L1} = (1 + C_3)/20$. The DDR RAM has a fixed hit time of 34 cycles per access, and never misses.

i) State the values of the constants $h_{L1}$ and $m_{L1}$, then give an expression for the Average Memory Access Time (AMAT) of the system that substitutes numerical values for all known constants.

ii) Use your knowledge of the typical relationships between L1 and L2 cache performance to provide a numeric lower-bound on the AMAT of this system. State the knowledge you rely on, and any other assumptions.

d The EEE department wishes to purchase a large number of new workstations for student use, and has received quotes from twelve potential suppliers. Each supplier's quote proposes a different system, and contains a variety of common quantitative information about the system's cost, performance, and capabilities.

i) Propose a quantitative metric which reduces each quote to a single number $m$, where lower values of $m$ are better. Your metric should rely on at least two clearly identified pieces of quantitative information from the quotes. Briefly justify why this metric makes sense from the point of view of purchasing workstations.

ii) Invent parameters for two different hypothetical systems, and use your metric to show which one should be bought.

iii) Give an example of a system which would do well according to your metric, but in fact would not be appropriate for the EEE department.

*The four parts carry, respectively, 20%, 20%, 30%, and 30% of the marks.*

2 A designer has decided to create a "compressed" 16-bit instruction format for MIPS. This format will be used in addition to the standard 32-bit instruction format, and binaries will be allowed to contain a mixture of 16-bit and 32-bit instructions. 32-bit instructions will be fetched and processed as normal. 16-bit instructions will be expanded into a full 32-bit instruction in the ID stage, with the EX and MEM stages will remain unchanged.

a What is the main advantage of supporting 16-bit instructions?

b Give an example of a 32-bit MIPS instruction which cannot be encoded as a single 16-bit instruction, regardless of the 16-bit encoding chosen.

c Describe the main changes needed to the instruction-fetch stage to support a mix of 16-bit and 32-bit instructions.

d Assume a program binary contains a sequence of three arithmetic instructions. The first and third instruction are 32-bit, while the second instruction is 16-bit, and the instructions are placed back-to-back in memory. Assuming the first and second instructions execute correctly, what problem will occur on the third instruction?

e It has been decided that the 16-bit J instruction will follow the form of the 32-bit J instruction, but with a reduced-size immediate field. What is the set of addresses a 16-bit J instruction can jump to? Explain any assumptions needed.

f Propose a new encoding for R-type instructions in the 16-bit format. Assume that there is still a 6-bit op-code, and the op-code has the same meaning as the normal 32-bit instruction op-code.

g Describe a tradeoff you had to make when designing your new R-type encoding.

*The seven parts carry, respectively, 15%, 10%, 15%, 10%, 15%, 20%, and 15% of the marks.*